R102 | Whole Matched A/B

Only TSP/ATSP donor initialization differs. BOTH routing donors are the identical pretrained R94B.

Arm / split	Top1 (%)	mean_cost	SBS	Oracle	vs_SBS (%)	Regret (%)
R102/A / TRAIN	71.7780	11.38686488	11.40070239	11.34507303	-0.2194	0.3719
R102/A / VAL	71.3012	11.41599712	11.42797847	11.37169190	-0.2004	0.3962
R102/A / TEST	71.5901	11.37211766	11.38529925	11.32993237	-0.2120	0.3784
R102/B / TRAIN	71.3612	11.38809538	11.40070239	11.34507303	-0.2064	0.3867
R102/B / VAL	71.1321	11.41632267	11.42797847	11.37169190	-0.1968	0.4008
R102/B / TEST	71.2259	11.37316519	11.38529925	11.32993237	-0.1999	0.3920

Arm / native LIB	n	Top1 (%)	mean_cost	vs_SBS (%)	Regret (%)
R102/A / TSPLIB	49	20.4082	8.21787372	-0.2738	0.9463
R102/A / CVRPLIB	100	65.0000	67.13353082	-0.1663	0.5098
R102/B / TSPLIB	49	22.4490	8.21366486	-0.3249	0.9165
R102/B / CVRPLIB	100	60.0000	67.15910571	-0.1283	0.5679

Same original seed2/R84 state + geometry, three fresh projections; routing projection seed940003.

A update 33120; B update 20160. Each is its own unique full18 minimum-VAL checkpoint.

Native/FP64 ind; TRAIN SBS; ALL is task-equal. No TEST/LIB tuning or per-task assembly.

R94B is historical, not matched. Source-pretraining compute/data overlap is not controlled.

Routing provenance identical: True; all18 metrics and NSS4 in arm PDFs.

R102/B versus R102/A | Fixed target36 versus matched R102/A

Native FP64 true-winner recovery; ALL deltas and cost contributions are task-equal.

Split	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	1851	183	993	330	2.2226
VAL	100	15	43	19	2.1622
TEST	92	14	57	23	2.0009

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.416735	21397	27473	0.001230500	0.014841
VAL	-0.169136	1296	1433	0.000325548	0.004669
TEST	-0.364198	1220	1515	0.001047530	0.013517

Split	New false-pick cost contribution	New false-pick cost DELTA contribution
TRAIN	0.002021721	0.000090268
VAL	0.002659335	0.000094697
TEST	0.002207645	0.000091395

False-pick costs use original FP64 selected costs / all instances, then equal-problem mean.

R84A and R94B are historical references, not matched controls. No TEST/LIB selection.

R102/A | Online Timing (all18)

GPU-resident batch128 amortized; one warmup + three synchronized repeats; donor hooks verified.

Problem	n	Donor + selector ms / instance	Cache-hit selector ms / instance
TSP	128	4.352492	0.265743
CVRP	128	9.159167	0.318541
ATSP	128	5.347474	0.069571
OVRP	128	3.680447	0.068022
VRPB	128	4.115417	0.068259
VRPL	128	4.184195	0.068375
VRPTW	128	4.037658	0.067818
OVRPTW	128	3.882628	0.068291
OVRPB	128	4.121012	0.068512
OVRPL	128	4.169456	0.068073
VRPBL	128	3.851259	0.068156
VRPBTW	128	4.212718	0.068230
VRPLTW	128	4.118054	0.068190
OVRPBL	128	4.191481	0.068259
OVRPBTW	128	4.031748	0.068147
OVRPLTW	128	3.959594	0.068167
VRPBLTW	128	3.789620	0.068200
OVRPBLTW	128	4.014874	0.067972

ALL18 include a real donor: TSP ELG128, ATSP row256+col256, routing frozen ReLD128.

Both nco_nodes and nco_node_mask removed each online call; hook counts4 online,0 cache.

Excludes dataset I/O, host transfer and one-time preparation; not singleton or end-to-end.

Cache-hit selector is NOT actual total inference. Device: cuda:0

R102/B | Online Timing (all18)

GPU-resident batch128 amortized; one warmup + three synchronized repeats; donor hooks verified.

Problem	n	Donor + selector ms / instance	Cache-hit selector ms / instance
TSP	128	4.545511	0.267213
CVRP	128	9.531808	0.322262
ATSP	128	5.604482	0.072253
OVRP	128	3.834003	0.070070
VRPB	128	4.298040	0.070430
VRPL	128	4.365036	0.070378
VRPTW	128	4.207879	0.070138
OVRPTW	128	4.053627	0.070489
OVRPB	128	4.279688	0.070190
OVRPL	128	4.363668	0.070480
VRPBL	128	4.024312	0.070726
VRPBTW	128	4.380677	0.070320
VRPLTW	128	4.302016	0.070421
OVRPBL	128	4.363686	0.070397
OVRPBTW	128	4.231044	0.070621
OVRPLTW	128	4.137682	0.071058
VRPBLTW	128	3.929782	0.070082
OVRPBLTW	128	4.219693	0.070415

ALL18 include a real donor: TSP ELG128, ATSP row256+col256, routing frozen ReLD128.

Both nco_nodes and nco_node_mask removed each online call; hook counts4 online,0 cache.

Excludes dataset I/O, host transfer and one-time preparation; not singleton or end-to-end.

Cache-hit selector is NOT actual total inference. Device: cuda:0